ECN 711
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Homework 6

1. **A version of a time-to-build model**
   (This problem appeared on a 2025 qualifier makeup exam)

   Consider a decision problem of an infinitely-lived agent in an economy in which time is discrete. The agent has access to the production technology that uses two types of capital – capital installed in the last period and capital installed two periods ago. That is, in period t, the output is

   $$y_t = A(\alpha k_{t-2} + (1-\alpha)k_{t-1})$$

   where $A > 1$, $\alpha \in [0, 1]$, and k_{t-2} and k_{t-1} are the capital levels installed in periods $t-2$ and $t-1$, respectively. The output in period t can be used for period-t consumption, c_t, and new capital installed in period t, $k_t \geq 0$ (which would be used in production in periods $t+1$ and $t+2$, and disappears thereafter).

   The agent's preferences are given by

   $$\sum_{t=0}^{\infty} \beta^t u(c_t), \quad \beta \in (0, 1),$$

   where $\beta \in (0, 1)$ and $u : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly increasing, strictly concave, and continuously-differentiable function, such that $\lim_{c \to 0} u'(c) = +\infty$.
   In period 0, the economy starts with the stocks of capital $k_{-1} > 0$ and $k_{-2} > 0$ installed in the two previous periods.

   (a) Set up the agent's problem recursively.
   (b) Assuming that the solution is interior, derive the Euler Equation and interpret it intuitively.
   (c) Assume that $u(c) = \ln(c)$.
   Note that if $\alpha = 0$, this becomes an optimal growth model with linear returns, and it is easy to verify that the optimal capital and output increase at a rate βA. That is,

   $$k^*_{t+1} = \beta A k^*_t \quad \text{and } y^*_{t+1} = \beta A y^*_t.$$

   Derive the closed-form solution for the path of optimal capital and output for the other 'extreme' case in which $\alpha = 1$. How does the capital and output growth rates in this economy compare to the growth rates in the economy with $\alpha = 0$? Provide a formal argument and an intuitive explanation.

2. **Optimal growth model with habit persistence**
   Consider an infinite horizon optimal growth model, in which per period utility exhibits "habit persistence": the utility derived from consuming amount c_t in period t also depends on the amount c_{t-1} that was consumed in the previous period. In other words, the agent's instantaneous utility can be written as $u(c_t, c_{t-1})$. The rest of the setup is the same as in the benchmark model studied in class (the net output in period t is given by $f(k_t)$ and the restriction $k_{t+1} \geq 0$ applies). Set up the agent's decision problem recursively. Carefully specify what the state and control variables are.

3. **Sequential Competitive Equilibrium with taxation**
   Consider an infinite horizon model populated by mass one of identical consumers, mass one of identical firms, and the government.

   In every period, the consumers have one unit of time endowment that they can split between leisure and working for hire. They also own a representative stock market portfolio that pays off dividends (which consumers take as given) in every period. Finally, they own capital stock which they rent to the firm in the beginning of each period. Capital depreciates at rate δ during the production process. The consumers' capital stock holding can be adjusted over time because the consumers decide in every period how much of their total revenue to consume, and how much to save as capital stock for the next period. The consumers' preferences are given by

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

   where $\beta \in (0, 1)$ and $u(c)$ satisfies all the standard assumptions, including the Inada conditions.

   The firms have access to production technology $F(k, n)$ is every period (also satisfying all the standard assumptions, including the Inada conditions), where k and n are capital and labor input, respectively. They can rent capital and hire labor on the competitive markets in order to maximize profits, and distribute all end-of-period profits to their shareholders in the form of dividends.

   There are three markets in every period: market for goods (where the prices are normalized to 1), market for labor (where labor services are traded at price w_t), and market for rental capital (where capital is rented at price r_t).

   The government needs to finance a stream of exogenous government expenditures $\{G_t\}_{t=0}^{\infty}$ (which it purchases from the goods market) using three types of taxes: lump-sum taxes T_t that apply to all the consumers equally regardless of their actions, proportionate tax s_t on consumption (meaning that is a consumer spend c_t on consumption, $s_t c_t$ is paid to the government as sales tax), and proportionate tax τ_t on capitals gains income (meaning that if a consumer earns $r_t k_t$ in rental income, (s)he needs to pay $\tau_t r_t k_t$ to the government). The government budget constraints is satisfied in every period, i.e.,

$$G_t = T_t + s_t c_t + \tau_t r_t^* k_t^*$$

must hold, where stars denote equilibrium values.

(a) (10 pts) Define the sequential trade competitive equilibrium in this economy. In the process, define all the necessary variables and decision problems.

(b) (10 pts) Derive the set of necessary and sufficient conditions that fully characterize the equilibrium.

(c) (10 pts) Set up the Social Planner's problem in this economy and derive the set of necessary and sufficient conditions characterizing the efficient allocation.

(d) (10 pts) Show that the equilibrium allocation is efficient if and only if $\tau_t = 0$ and $s_t = s_{t+1}$ for all $t \geq 0$.

(e) (5 pts) It appears, a constant sales tax rate in this economy is not distortionary. Can you think of a simple (and familiar to you) modification of this model in which any positive consumption tax would be distortionary?